Answers:

I. 1. (b) 2. (b) 3. (c) 4. (c) 5. (c) 6. (a) 7. (d) 8. (a) 9. (b)

10. (a) 11. (b) 12. (d) 13. (b) 14. (a) 15. (c)

II. 16. 1 17. $\frac{1}{2^{12}}$ 18. Independent 19. $\frac{(b-a)^2}{12}$ 20. N (0, 1)

III. 24. $10C_2 0.32^2\ 0.68^8$ 25. Mean = 1280 SD = 16 27. 0.0126

29. Mean $\lambda = 2$ 31. Mean = 0; Variance = $\frac{100}{3}$ 34. 0.9973

IV. 37. (i) $0.9^{18} \times 1.90$ (ii) $0.9^{20} \times 7.1317$ (iii) $1\text{-}0.9^{20} \times 3.222$

38. (i) 0.2^{10} (ii) $(0.8)^{10}(8.19)$ 39. 0.5595 40. (i) 0.2240 (ii) 0.1847

41. (i) 16 (ii) 16

V. 43. (i) $p(x) = 4C_x\ (0.2575)^x\ (0.7425)^{4-x}$ (ii) 62, 84, 45,10, 1

44. (i) $p(x) = e^{-0.25}\ \frac{0.25^x}{x!}$ (ii) 78, 20, 2, 0, 0

45. (i) 0.05395 (ii) 0.05498 46. Mean = 101.12 Standard deviation = 20.65

47. (141.589, 158.411)

48. $p = 0.53125$

Number of Heads	0	1	2	3	4
Expected frequency	3	14	24	18	5

49. (i) 0.5168513, (ii) 0.1125702, (iii) 0.99530888, (iv) 0.990147

50. (i) Rohit, Rahim (ii) Pavithra, Priya, Fazeena

(iii)

Student	Rohit	Pavithra	David	Rahim	Priya	Fazeena
Mark	172	110	150	184	118	134